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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	JONG JAVA
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4 (primary), SDG 8 and SDG 10
Project / Solution Name	JOBRIDGE
Name (Leader)	Louis Benedict Oen
Email & Phone (Leader)	**louisbenedict127@gmail.com , +60 11 2371 3971**
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5 members

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In this modern era, AI has transformed the way humans work, while new jobs continue to emerge, some existing roles are now at risk of being replaced by technology. According to the International Labour Organization, administrative positions are the easiest jobs to be replaced by AI with women facing twice the risk compared to men. In Malaysia, TalentCorp (2024) found that 620,000 jobs could be highly affected by AI, digitalisation, and the green economy.

This situation presents a challenge for several groups, such as fresh graduates who struggle to find employment that aligns with their skills. Meanwhile, some current employees fear that their jobs might be replaced by AI and those looking to change professions often feel uncertain, worrying that their existing skills may not be applicable to future roles. Consequently, they need to understand which jobs and skills are important for survival in the future, and also how they can acquire these skills affordably.

On the other hand, companies face difficulties in searching for the right candidates and choosing which skills their employees need to develop. This makes it challenging for job seekers to secure suitable employment, while companies struggle to find the right talent. This issue cannot be taken lightly, if left to continue, it will slow down economic growth, that will slow things down the achievement of Sustainable Development Goal (SDG) 8 (Decent Work and Economic Growth) and SDG 10 (Reduced Inequalities).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

JOBRIDGE is a live web app that transforms real-time job market data into a clear, personalized career path. It helps you navigate from a job with high generative AI exposure to a realistic, safer alternative.

a. **Future Skills Radar:** We collect monthly job postings from Hacker News (since Jan 2020) to track AI adoption and demand for 128 skills. Using the best-tested forecasting model, the app projects skill needs 12 months ahead under both moderate and fast AI growth scenarios.
b. **CV Intake:** Upload or paste your CV. Skills are recognized transparently alongside the words that triggered them, so you can confirm them easily. To ensure fairness, names, genders, and ages are automatically removed before scoring.
c. **Explainable Matching:** Using LinkedIn job profiles, each role is scored by how much of its future skill demand you already cover. It also shows your exact shift in AI exposure (e.g., Gradient 4 to Gradient 2).
d. **Reskilling Pathway:** Missing skills are turned into a step-by-step learning plan using 127 verified free or low-cost courses, showing your expected job match score once completed.
e. **SDG Links & Team View:** Roles and skills connect to SDG targets, letting you personalize rankings by your goals. HR teams also get a skills gap heatmap for their workforce.

From SDC Phase 1 workshops, we gained several tools and technical skills that are very useful for our projects. For instance, Replit that we used for collecting useful codes and improving our model and Runpod, where we can run our model and evaluate to improve our model.

3. Technical Architecture & Implementation Plan

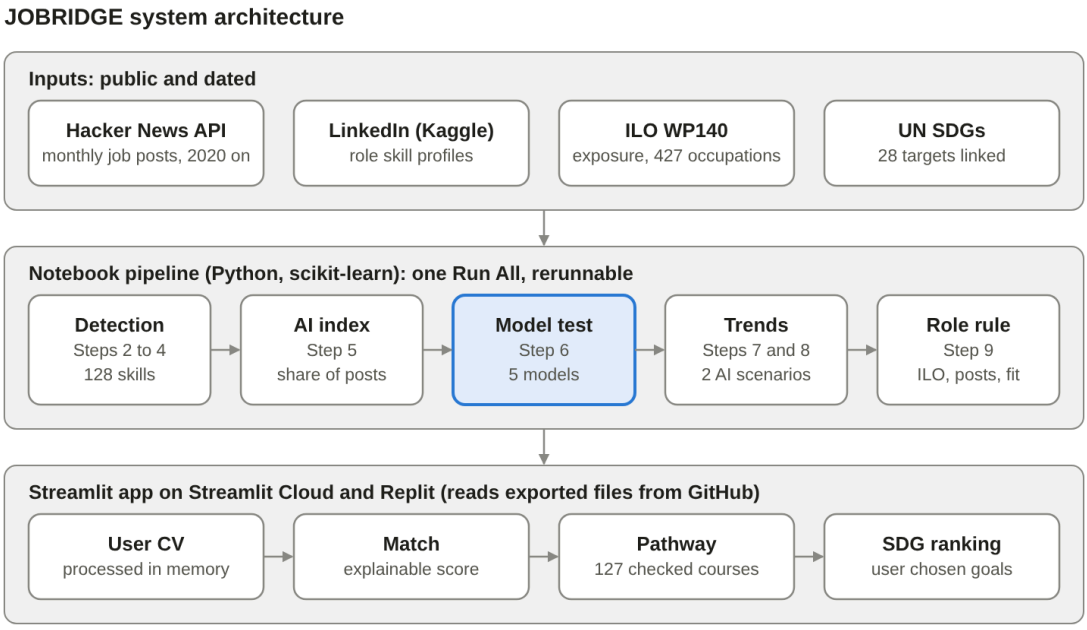
3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

a. Front end and back end: Flask web app with Plotly charts, CV upload as text or PDF (pypdf).

b. Data and machine learning: Python, pandas and NumPy, scikit-learn (ridge regression, histogram gradient boosting) for the forecasting experiment, one rerunnable Jupyter notebook builds every dataset.
c. Data sources and APIs: Hacker News Algolia API (live, monthly job posts), LinkedIn Job Postings 2023 to 2024 (Kaggle), ILO Working Paper 140 exposure index (427 occupations), UN SDG targets, a catalogue of 127 checked courses.
d. Storage: versioned CSV and JSON files in the GitHub repository, raw job text is never stored. A database is not needed at this scale and can be added in Phase 2 for user accounts and progress tracking.
e. Deployment: GitHub repository, deployed to Streamlit Community Cloud (public URL) and Replit.
f. System block diagram: public data feeds the notebook pipeline (skill detection, AI index, model experiment, trends, role rule), its exported files are committed to GitHub, the app reads only those files and takes the user from CV to match, pathway and SDG ranking.



g. Sponsor platforms: Yes. Replit, provided through the SDC programme, hosts a second deployment imported from our GitHub repository, set up with help from Replit Agent. No sponsor GenAI API is called at run time, the app uses no language model.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

a. Empathise: we started from admin, finance, payroll and contact centre workers, whose occupations the ILO rates Gradient 3 or 4, and from employers who want to redeploy them. They need realistic next steps, free courses and reasons they can trust.
b. Define and ideate: "Show me a job I can reach that AI is less likely to take over, and what to learn first." Our first version recommended data roles; we dropped that because the jump is too big, and the least exposed jobs are mostly niche or senior. Instead, we set a measurable rule, which is ILO Gradient 2 or lower, at least 50 job posts, and at least 10% skill overlap with admin and finance work.
c. Prototype and test: a working app built from one rerunnable notebook that is checked by 69 automated detection tests, a rolling backtest of five forecasting models and a twin CV fairness test (CVs differing only in name, gender and age get identical results).
d. Deploy: the app is live on Replit, refreshing the data, so trends update monthly (when we push in the project repository). In Phase 2 we will test with admin and finance workers and add Malaysian job data.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary: SDG 4.4** Substantially increases the number of youth and adults who have relevant skills for employment, decent jobs and entrepreneurship. **Secondary: SDG 8.5** Full and productive employment and decent work for all women and men; and **SDG 10.2** inclusion of all irrespective of age or sex, supported by our twin CV fairness test.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	620,000 jobs, 18% of the 3.5 million employees in 10 key Malaysian sectors, are classed as highly impacted by AI, digitalisation and the green economy (TalentCorp study, April to September 2024). Globally, clerical occupations have the highest generative AI exposure; 3.3% of all employment is in the highest exposure gradient, 4.7% for women versus 2.4% for men (ILO Working Paper 140, 2025). Our starting roles are all ILO Gradient 4 or 3: accounting and bookkeeping clerks 0.64, payroll clerks 0.61, general office clerks 0.60, contact centre clerks 0.58. **Sources:** Vulcan Post, 21 November 2024, https://vulcanpost.com/876459/talentcorp-workforce-study-job-sectors-risk-ai-digitalisation-malaysia; ILO (2025), Generative AI and Jobs: A Refined Global Index of Occupational Exposure, Working Paper 140, https://www.ilo.org/publications/generative-ai-and-jobs-refined-global-index-occupational-exposure

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary:** projected skills match gain, in percentage points, towards a recommended role that is less exposed to generative AI after the user completes the JOBRIDGE pathway (computed in the app for every CV). **Supporting:** pathway coverage, the share of a user's top 6 skill gaps that have a checked free or low cost course; and the share of pilot users who rate their top recommendation as realistic.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	1. An average projected match gain of at least 20 percentage points across our sample personas and pilot users, with every recommended move role at least one ILO gradient less exposed than the user's current role. 2. Pathway coverage raised from 73% of the 128 tracked skills today (93 skills with a checked course) to at least 90%. 3. At least 80% of pilot users from admin and finance roles rate their top recommendation as realistic.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Flask	Code	Apache License 2.0
pandas and NumPy	Code	BSD 3 Clause License
scikit-learn	Code	BSD 3 Clause License
Plotly (plotly.py)	Code	MIT License
matplotlib	Code	Matplotlib License (PSF based)
pypdf	Code	BSD 3 Clause License
requests	Code	Apache License 2.0
Hacker News “Who is hiring?” threads via HN	Dataset	Public API; only aggregate statistics stored, each post

Algolia API		linked to its source
LinkedIn Job Postings 2023 to 2024 (Kaggle, arshkon)	Dataset	Kaggle dataset licence (see dataset page); not redistributed
ILO Working Paper 140 (2025), Table A1	Dataset	© ILO 2025; cited with page references
UN SDG goals and targets	Dataset	United Nations; cited
Course catalogue (127 provider pages)	Dataset	Links and checked facts only; content remains with each provider
Claude (Anthropic)	Model	Anthropic terms of service; coding assistant during development, not used in the app
Replit Agent	Model	Replit terms of service; deployment setup

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon.

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Louis Benedict Oen]

[24-10-2026]